

Lec-03

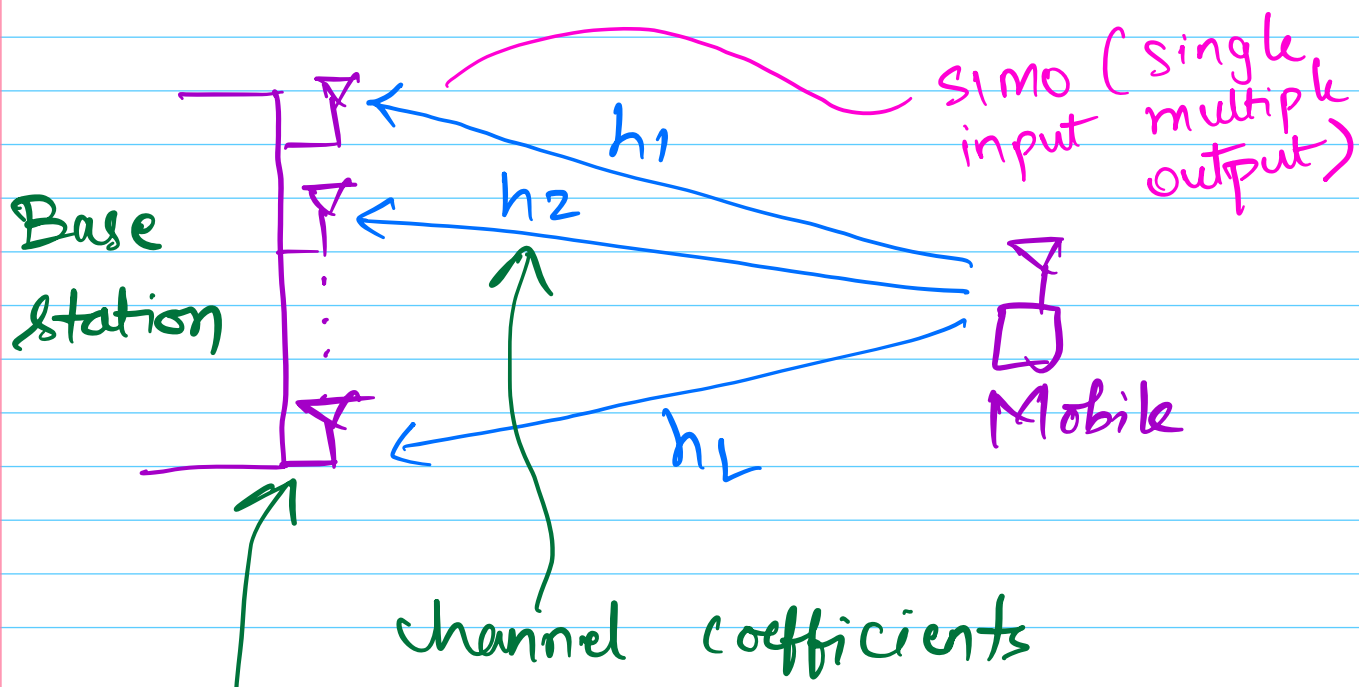
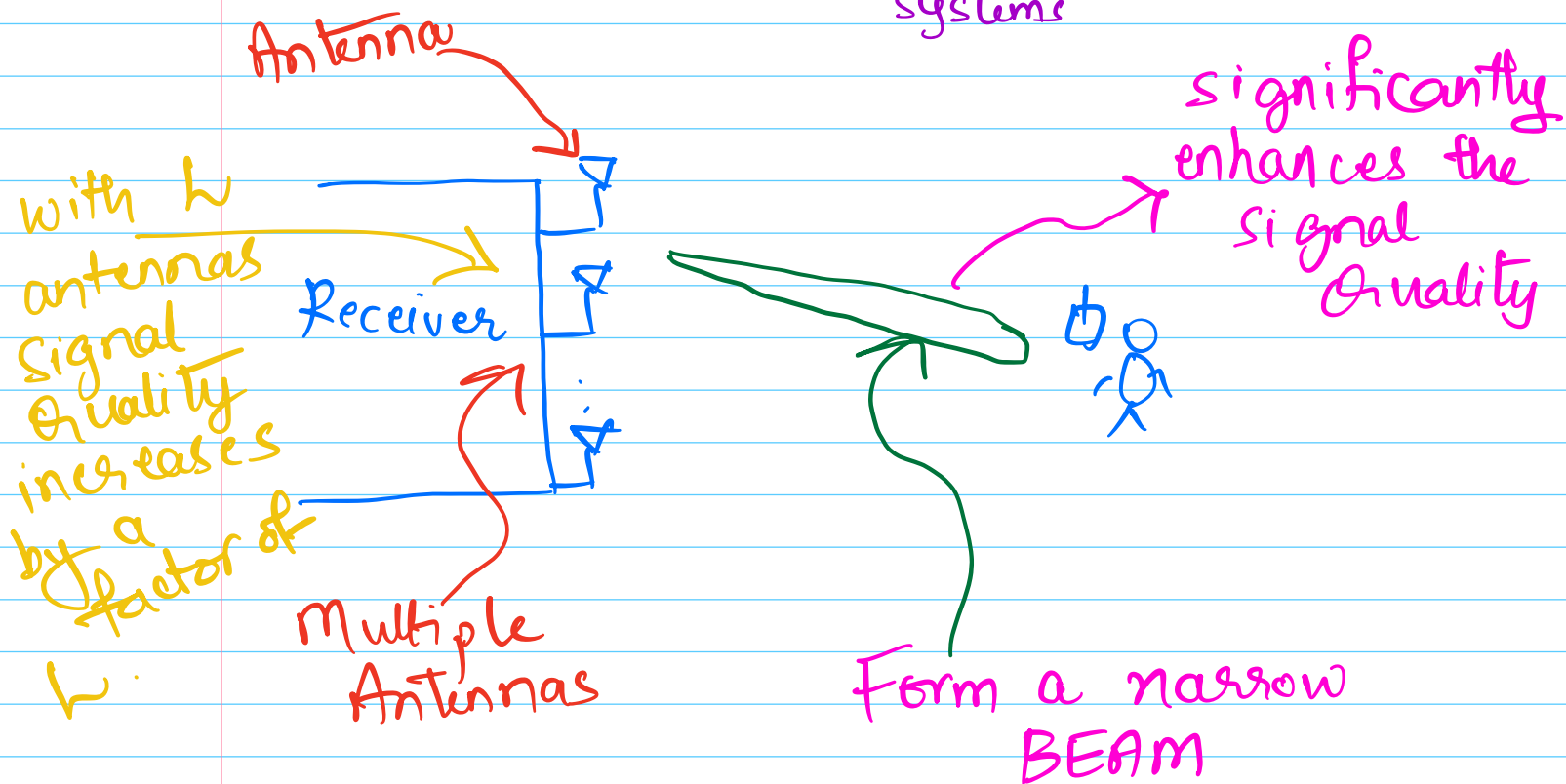
Vector Properties

Very very important!

4G/5G/WiFi

BEAM FORMING

in Wireless systems



L antennas $h_i =$ Fading channel coefficients

$y_i =$ Received symbol or
output symbol on
antenna i .

$x =$ Transmitted symbol.

$$y_i = h_i x + n_i$$

input/output
i/o model for antenna i

Noise on
antenna i .

$$\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_L \end{bmatrix} = \begin{bmatrix} h_1 \\ h_2 \\ \vdots \\ h_L \end{bmatrix} x + \begin{bmatrix} n_1 \\ n_2 \\ \vdots \\ n_L \end{bmatrix}$$

models for L antennas

$$\underbrace{\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_L \end{bmatrix}}_{\bar{y}} = \underbrace{\begin{bmatrix} h_1 \\ h_2 \\ \vdots \\ h_L \end{bmatrix}}_{\bar{h}} x + \underbrace{\begin{bmatrix} n_1 \\ n_2 \\ \vdots \\ n_L \end{bmatrix}}_{\bar{n}}$$

$$\bar{y} = \bar{h} x + \bar{n}$$

output vector channel vector noise vector

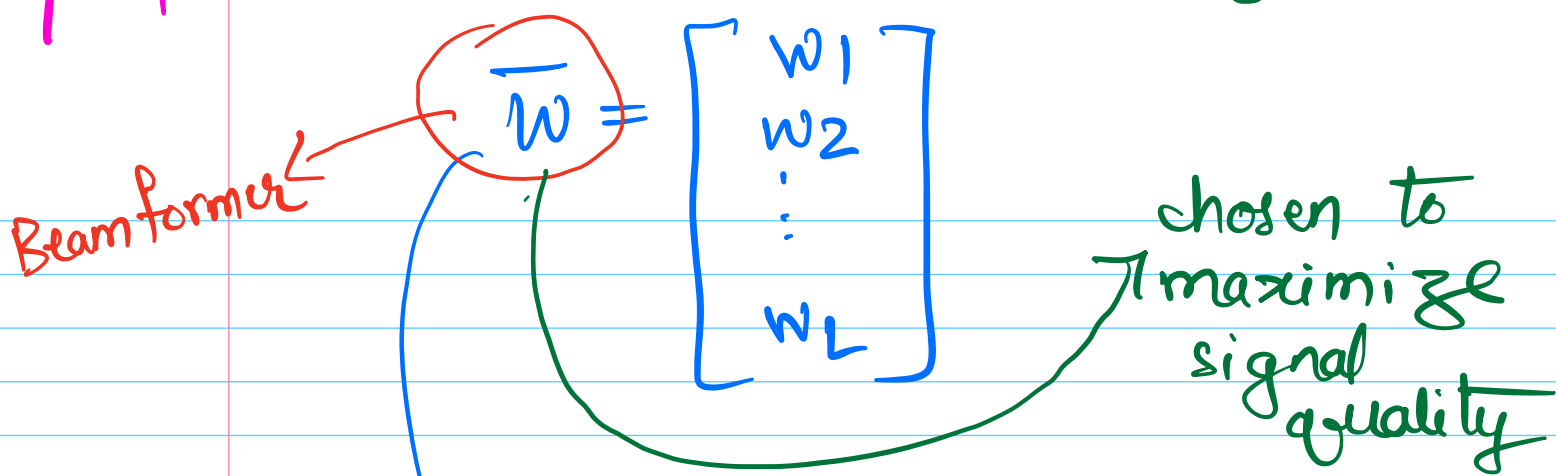
Beamforming

perform weighted linear combination

$$w_1^* y_1 + w_2^* y_2 + \dots + w_L^* y_L$$

$$= \underbrace{\begin{bmatrix} w_1^* & w_2^* & \dots & w_L^* \end{bmatrix}}_{\bar{w}^H} \underbrace{\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_L \end{bmatrix}}_{\bar{y}}$$

inner product of
Beamformer
input = $\bar{w}^H \bar{y}$



Transpose & Conjugate

$$\bar{\mathbf{w}}^H = [w_1^* \ w_2^* \ \dots \ w_L^*]$$

How to choose $\bar{\mathbf{w}}$?

Optimal $\bar{\mathbf{w}}$

That maximizes SNR (Signal to Noise Ratio)

is $\bar{\mathbf{w}} = \frac{\bar{\mathbf{h}}}{\|\bar{\mathbf{h}}\|}$

maximizes SNR
Maximal Ratio
Combiner (MRC)

unit Norm
vector

matched filter